

BITS Pilani

Pilani|Dubai|Goa|Hyderabad

# BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

## Work Integrated Learning Programmes (WILP) Division

## Dissertation Orientation Session for M.Tech DSE / M.Tech AIML/CSIS

## 1st Semester 2026-27

# Schedule

innovate achieve lead

| NSP 2 S1-26 Dissertation Calendar | Semester Start -24th October 2026<br>Semester End - 5th March 2027 |
|---|---|
| | |
| Dissertation General Orientation | 1st August 2026,Saturday 7:30 pm to 9:00 PM IST |
| Dissertation Domain Orientation | 9th August 2026 Sunday 8:00 pm to 8:30pm IST |
| | |
| Submission of Student' Project Details on the Elearn Taxila Portal | 21st September to 20th October 2026 |
| Access of Viva portal to students with time lines | 24th October 2026 |
| Submission of outline/ Abstract Report | 31st October to 7th November 2026 |
| Student VIVA date selection | 12th November to 15th November 2026 |
| Abstract VIVA & Grading Evaluation by BITS Faculty Panel | 20th November to 4th December 2026 |
| | |
| Plagiarism Check Enablement | 13th December 2026 |
| Mid-Semester Report submission | 18th December to 25th December 2026 |
| Student VIVA date selection | 30th December 2026 to 2nd January 2027 |
| Mid Semester VIVA & Grading Evaluation by BITS Faculty Panel | 7th January to 21st January 2027 |
| | |
| Plagiarism Check Enablement | 26th January 2027 |
| Final Dissertation | 31st January to 7th February 2027 |
| Student VIVA date selection | 12th February to 15th February 2027 |
| Final Viva to be conducted by the BITS examiner | 20th February to 5th March 2027 |
| | |

# Dr. Chetana Gavankar

- 28+ Years of Experience in IT, Teaching and Research
- Working as Professor and Program Lead in the CSIS department at Work Integrated Learning Programmes Division, BITS Pilani.
- PhD in Computer Science and Engineering from a joint programme of IIT Bombay and Monash University, Australia.
- Working on different state of art research projects and have been awarded the “Best Industry Aligned Research” at the CSI TechNext India - Awards to Academia.
- Published various papers and is also a reviewer at national and international level peer reviewed conferences and journals

# Content

- Defining the scope of your dissertation
- AIML project areas and applications
- AIML domain-wise project examples
- DSE project areas and applications
- DSE domain-wise project areas
- Software systems project topics
- Software systems domain-wise project areas
- Q & A session

# Defining the Scope



- Focus on specific industry problem
- Define clear research question
- Select relevant dataset/domain
- Identify a research gap from literature (try to look at recent work)
- Choose core technical approach
- Define measurable evaluation metrics
- Build a working prototype
- Keep scope manageable: 1 problem + 1 dataset + 1 approach + 3–5 experiments
- Remember: an assignment applies known solutions to a given problem; a dissertation defines the problem and defends a contribution with evidence
- Justify your choices: why this approach, why this tool stack, why these evaluation measures — and what alternatives you rejected

# AIML Project Areas

- Agentic AI with tool use
- RAG and Graph-RAG enterprise assistants
- Small language models: fine-tuning and distillation
- Multimodal AI (text, image, audio, video)
- Document intelligence (contracts, invoices, claims)
- Multilingual and Indian language processing
- Healthcare AI: diagnostics and clinical documentation
- Speech AI and real-time voice agents
- Computer vision for smart manufacturing
- Reinforcement learning and model alignment
- Explainable, safe and trustworthy AI
- LLMOps: evaluation, guardrails and observability
- Efficient inference and edge deployment
- Personalized recommendation and semantic search
- Federated and privacy-preserving learning

# Artificial Intelligence & Machine Learning applications



**Agentic AI & Autonomous Systems:**
Collaborating agents that plan and execute multi-step tasks with tools, memory and human oversight, e.g. coordinating a supply chain.

**Domain Small Language Models (SLMs):**
Fine-tune and distill compact models for legal, banking or regional Indian languages, deployable on-premise or on-device.

**RAG & Graph-RAG for Enterprise Knowledge:**
Retrieval over documents and knowledge graphs with grounded citations and evaluation to control hallucination.

**Explainable & Responsible AI for Fintech:**
Dashboards that explain loan or underwriting decisions and audit models for bias and compliance.

**Federated Learning for Healthcare:**
Let hospitals jointly train a diagnostic model (e.g., diabetic retinopathy) without sharing patient data.

**Reinforcement Learning for Robotics & Alignment:**
Train robotic manipulation policies, or apply preference-based post-training to align model behaviour.

**Multimodal Understanding:**
Combine vision, speech and text for real-time emotion recognition, video analytics or assistive applications.

# AIML Domain-wise Project Examples



**Healthcare:**
Clinical note summarization, medical imaging triage, federated diagnostic models.

**Banking & Fintech:**
Fraud detection agents, credit risk scoring with explainability, KYC document intelligence.

**Retail & E-commerce:**
Conversational shopping assistants, demand forecasting, visual search and recommendations.

**Manufacturing & Energy:**
Visual defect inspection, predictive maintenance, RL-based process optimization.

**Telecom & IT Operations:**
Network anomaly detection, AIOps incident triage, automated support copilots.

**Education & Public Services:**
Multilingual tutoring assistants, grievance classification, document Q&A over policy archives.

**Mobility & Logistics:**
Route and fleet optimization, driver safety vision systems, warehouse robotics.

# DSE Project Areas

- Real-time streaming pipelines with Kafka and Flink
- Lakehouse architecture and open table formats (Iceberg, Delta)
- ETL to ELT modernization with dbt and cloud warehouses
- Vector databases and retrieval infrastructure for AI apps
- Feature stores for real-time model inference
- Time-series forecasting for demand, energy and sales
- Fraud and anomaly detection with graph analytics
- Data quality, drift detection and observability frameworks
- MLOps and LLMOps: deployment, versioning and monitoring
- Data governance, lineage and privacy-preserving analytics
- IoT and edge analytics with on-device filtering
- Natural language interfaces to enterprise data

# Data Science & Engineering applications

**Real-time Analytics on Streaming Data:**
Build a scalable Kafka and Flink pipeline to analyze live public transport or payments data.

**Lakehouse & Open Table Formats:**
Design a lakehouse on Iceberg or Delta with ELT transformations and incremental modelling.

**Data Infrastructure for AI Applications:**
Engineer vector search and retrieval pipelines that serve embeddings for RAG workloads at scale.

**MLOps & LLMOps for Continuous Deployment:**
Automate testing, deployment, evaluation and monitoring of models and LLM-powered services.

**Graph Neural Networks for Fraud Detection:**
Implement a GNN to identify fraudulent transactions within the UPI network.

**AI for Sustainable Agriculture:**
Use satellite imagery and sensor data to predict soil health and optimize irrigation.

**Data Quality, Drift & Governance:**
Monitor production data for quality issues, drift and lineage, triggering alerts or retraining.

# DSE Domain-wise Project Areas

**Banking & Fintech:**
Real-time payment fraud pipelines, risk data marts, regulatory reporting automation.

**Healthcare:**
Clinical data lakehouse, patient 360 views, privacy-preserving analytics on EHR data.

**Retail & E-commerce:**
Clickstream analytics, demand forecasting, real-time personalization feature stores.

**Manufacturing & Energy:**
IoT sensor pipelines, predictive maintenance analytics, edge data filtering.

**Telecom & IT Operations:**
Network telemetry processing at scale, AIOps observability, churn analytics.

**Logistics & Mobility:**
Fleet and route optimization data platforms, ETA prediction, supply chain visibility.

**Public Sector & Education:**
Open data platforms, learning analytics dashboards, data governance and lineage frameworks.

# Software Systems Project Areas

- AI-Based Software Bug Prediction Using Machine Learning
- LLM-Based Automated Code Review and Vulnerability Detection
- Cloud-Native Application Performance Optimization Using Kubernetes
- AI-Driven Software Testing and Test-Case Generation
- Microservices-Based Systems: Performance and Scalability Analysis
- Blockchain-Based Secure Data Management for Distributed Systems
- Anomaly Detection in Cloud Infrastructure Using Machine Learning
- Privacy-Preserving Machine Learning for Sensitive Data

# Software Systems Domain-wise Project Areas



**Software Engineering & Quality:**
AI-assisted code review, automated test generation, defect and bug prediction.

**DevOps & Platform Engineering:**
CI/CD automation, infrastructure-as-code validation, release risk prediction.

**Cloud & Distributed Systems:**
Microservices observability, autoscaling policies, cost and performance optimization.

**Cybersecurity:**
Intrusion and malware detection, secure code analysis, threat intelligence correlation.

**Enterprise Applications:**
Legacy modernization and refactoring, workflow automation, API integration platforms.

**IoT & Embedded Systems:**
Edge device management, firmware update pipelines, real-time control software.

**Blockchain & Digital Trust:**
Smart contract verification, distributed ledger applications, identity and access systems.

# Q & A Session

# Thank You